

From Extreme Samples to Condition-Monitoring Evidence: A Five-Dataset Dynamical-EVT Diagnostic Workflow for Predictive Maintenance

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Abstract

Industrial predictive-maintenance studies often compress heterogeneous monitoring problems into pointwise anomaly scores. That framing loses the distinction between telemetry samples, alarm episodes, repair histories, component-condition cycles, and production-quality outcomes. This paper reports a reproducible dynamical-EVT diagnostic workflow evaluated through a five-dataset evidence matrix spanning MetroPT, MetroPT2, SCANIA Component X, UCI Hydraulic Systems, and UCI SECOM. The implementation combines causal state construction, regime-conditioned thresholds, declustering, extremal-index diagnostics, event-level matching, conservative tabular condition diagnostics, baseline provenance, and a frozen evaluation protocol. The completed asset bundle records 7 claim-ledger entries, of which 7 are permitted for manuscript use under the generated ledger. The present evidence supports a bounded conclusion: the workflow makes dependence, clustering, alarm burden, calibration diagnostics, estimand changes, and weak real-data results visible across several public industrial-monitoring angles. It does not support a population-level superiority claim beyond the registered evidence matrix. All reported claims are linked to experiment identifier 0b597ad010969c41, protocol hash 96461a9ff099d0d7, and generated artifacts.

1 Introduction

Industrial maintenance alarms are useful only when they map to the decision unit that operators and engineers actually face. In some settings the unit is a failure episode, in others a repair history, a laboratory component-condition cycle, or a production example with a downstream yield label. Standard pointwise anomaly detection can be accurate at the sample level while still producing a poor operational object. A burst of extreme samples near one physical failure may be counted as many alarms; a vehicle-level repair problem may be misread as timestamp-level warning; and millions of telemetry rows can obscure the small number of independent events or entities that actually support inference.

Cyclic machinery creates an additional statistical problem. Compressor, engine, and actuator signals are shaped by operating regimes, control loops, maintenance states, and sensor preprocessing. Extreme observations therefore need not be independent exceedances of a stationary distribution. Dynamical extreme-value theory connects high observations of functions evaluated along a trajectory with returns to small target sets, and it explains why periodic or recurrent structure can create clustered extremes [4, 5]. Under finite noisy observations, the same asymptotic quantities may be masked or distorted [2]. These facts make naive threshold exceedance counts an unstable basis for maintenance claims.

This paper studies a constrained contribution: a reproducible diagnostic workflow that turns machine-state trajectories and tabular condition-monitoring examples into artifact-backed evidence, with explicit warnings about what the artifacts do not prove. The workflow defines causal

state vectors, operating regimes, training-defined dangerous regions, regime-conditioned thresholds, exceedance clusters, lagged multivariate patterns, alarm episodes, and dataset-specific condition diagnostics evaluated under a frozen protocol. The central question is not whether a new model is a general winner across industrial settings. The question is whether generated artifacts make dependence, clustering, alarm burden, sensitivity, calibration, estimand changes, and unsupported claims visible enough to constrain a defensible predictive-maintenance paper.

The contributions are therefore deliberately bounded. First, the implementation provides a dynamical-EVT-inspired formulation for regime-conditioned extreme diagnostics in predictive maintenance. Second, it produces finite-sample simulation and sensitivity artifacts for extremal-index recovery and threshold/run-length choices. Third, it runs a five-dataset real-data matrix covering compressor telemetry, fleet repair histories, hydraulic component-condition cycles, and semiconductor process/yield examples. Fourth, it freezes evaluation and audit protocols so recall, false alarms, duplicate alarms, lead time, calibration, verification status, and industrial diagnostic rows are generated rather than hand-curated. Finally, it generates a claim ledger that allows only artifact-backed statements in the manuscript and provides build commands that regenerate the figures, tables, provenance, manuscript, supplement, and audits from version-controlled code. These contributions correspond to claim-ledger entries CLM-001 through CLM-007; claims outside that ledger are not used as positive evidence.

2 Related Work

Predictive-maintenance models often begin with supervised classification, reconstruction error, distance-based anomaly detection, or streaming thresholds. These methods are useful engineering baselines, but their common output is a timestamped score rather than an episode-level maintenance decision. EVT-based streaming detectors such as SPOT use peaks-over-threshold modelling to adapt anomaly thresholds in streams [9]; they address threshold calibration, but they do not by themselves resolve operating-regime mixture, duplicate alarms per failure, or the distinction between clustered observations and independent maintenance events.

Classical declustering and extremal-index estimation address dependence among extremes. Ferro and Segers proposed an inference scheme for clusters of extreme values using inter-exceedance times and the extremal index [3]. In dynamical systems, hitting-time statistics connect returns to target neighborhoods with extreme-value laws [4]. Periodic points can generate an extremal index smaller than one, giving a mathematical route from recurrence to clustered extremes [5]. The finite-sample industrial setting is less clean: noise may make deterministic recurrence appear closer to independence, and observed clustering can also arise from smoothing, missingness, persistence, or regime mixture [2]. Recent multivariate work extends dynamical EVT to cross-sectional dependence among vector-valued observables [1], which motivates the lag-pattern diagnostics used here.

The industrial datasets used by the repository are public condition-monitoring resources with different independence units and labels. MetroPT is an air-production-unit telemetry dataset for predictive maintenance [10]. MetroPT2 is a replication-style benchmark released through Zenodo [11]. SCANIA Component X is a fleet-level multivariate time-series dataset with operational histories and repair records for an anonymized component [7]. UCI Hydraulic Systems provides laboratory hydraulic test-rig cycles with component-condition targets [6]. UCI SECOM provides semiconductor process measurements with pass/fail yield labels [8]. These datasets broaden the empirical base, but they do not remove the need for estimand-specific accounting: one long compressor fault interval, one truck repair history, one hydraulic load cycle, and one wafer-level yield example are not interchangeable statistical units. The methodological gap addressed here is the absence of a reproducible layer that connects dynamical extreme diagnostics to alarm episodes, tabular condition diagnostics, sensitivity analyses, and claim constraints.

3 Method

Let $Z_t \in \mathbb{R}^p$ denote the causal machine-state vector observed at time t . The state may contain raw sensors, engineered lag features, or delay embeddings, but every coordinate must be available at or before t . Let R_t denote the operating regime, obtained either from rule-based compressor states, change-point segmentation, or a hidden-state model. Regimes are not interpreted as physical causes by themselves; they are conditioning variables used to reduce mixture effects when estimating tail and alarm behavior.

A dangerous region A is defined from training data only. In the implementation, A may be represented by failure-proximal prototypes, robust high-risk state summaries, or distance constraints. For a distance $d(\cdot, A)$, the scalar observable is

$$X_t = -\log\{d(Z_t, A) + \epsilon\},$$

where $\epsilon > 0$ prevents numerical singularities. Large X_t corresponds to a state closer to the training-defined target region. This transformation is a diagnostic construction, not a proof that the machine is mechanically degrading.

For each regime r , a threshold $u_r(q)$ is fitted from the training or calibration partition at quantile q . Exceedances are $I_t(q, r) = \mathbf{1}\{X_t > u_r(q), R_t = r\}$. Consecutive exceedances are converted into clusters using a run length m : two exceedance samples are assigned to the same cluster when the gap between them is no larger than m . The runs extremal-index estimate summarizes clustering under the selected q and m , while the inter-exceedance estimator provides an alternative declustering view [3]. The inverse extremal index is discussed only as an asymptotic cluster-size quantity under stated assumptions.

The hitting-time target for a warning horizon h is $Y_t^{(h)} = \mathbf{1}\{\tau_A(t) \leq h\}$, where $\tau_A(t)$ is the first future time at which the trajectory enters the target region or a labelled failure episode. A horizon-risk score estimates $\Pr(Y_t^{(h)} = 1 \mid \mathcal{F}_t)$ using causal features. Multivariate lag patterns are represented by tuples (j, k, ℓ) indicating that component j exceeds its threshold and component k exceeds its threshold after lag ℓ . These patterns are evaluated against null models that preserve marginal exceedance rates or within-cluster structure.

Pointwise alarms are finally merged into alarm episodes using the frozen merge gap and matched to failure episodes under the declared early-warning policy. This separation is essential: observations, exceedances, clusters, alarm episodes, failure episodes, repairs, and independent units are different objects. The evaluation metrics in Section 6 are therefore computed after episode conversion rather than directly from timestamp labels.

Not every registered dataset exposes event-onset labels. For Hydraulic Systems and SECOM, the real-data matrix therefore uses a conservative condition-diagnostic adapter rather than the compressor event metric. Sensor traces or process features are converted into causal feature vectors, a robust training-split center and scale are fitted, and test examples above the registered high quantile are compared with the dataset-specific target. The resulting precision, recall, F_1 , Brier score, target count, prediction count, and limitation text are reported as condition-monitoring diagnostics only. They broaden the study base across industrial angles, but they are not treated as event-level early-warning replications.

4 Simulation Study

The simulation layer is designed to falsify overconfident interpretations before industrial data are used. The registered systems include independent extremes, periodic dynamics, noisy periodic dynamics, regime mixtures, missingness perturbations, and degradation-like cyclic trajectories. Each simulation cell records a deterministic experiment identifier, configuration hash, random seed, threshold grid, run-length grid, estimator status, and raw outputs sufficient to regenerate the paper assets.

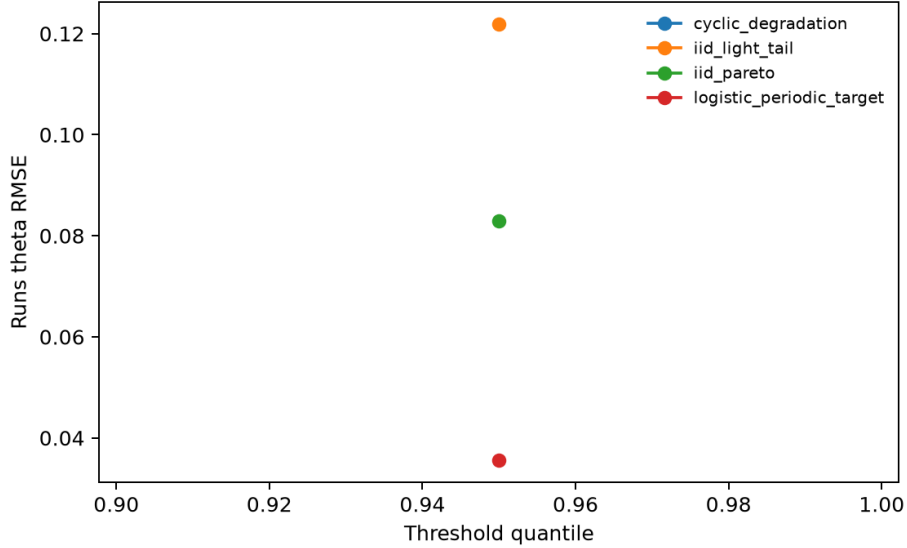


Figure 1: Simulation recovery summary generated from the registered simulation-study artifact. The plotted quantity is runs-estimator RMSE by threshold quantile and simulation family.

The main estimands are finite-sample bias and root mean squared error for extremal-index estimators, coverage or failure status where intervals are available, and recovery of clustered episodes. The simulation artifacts deliberately preserve estimator failures. A failed fit is treated as a terminal experiment outcome rather than an infrastructure error, because failure regions are part of the statistical evidence.

Figure 1 and the generated simulation table report the completed smoke-scale recovery artifact. The values support claim CLM-003: the controlled simulation output quantifies recovery error across threshold settings. They do not support a statement that the same estimator is reliable for every industrial fault interval.

5 Datasets

The repository registry covers MetroPT, MetroPT2, SCANIA Component X, Hydraulic Systems, and SECOM. MetroPT contains air-production-unit telemetry from an urban metro setting with pressure, temperature, current, digital control signals, and documented maintenance/failure intervals [10]. MetroPT2 provides a benchmark replication release with documented failure periods [11]. SCANIA Component X is a fleet-level dataset containing anonymized operational variables, repair records, and truck specifications for a proprietary component [7]. Hydraulic Systems adds cycle-level component-condition measurements from an experimental hydraulic test rig [6]. SECOM adds high-dimensional semiconductor process measurements with pass/fail yield labels [8].

The paper assets used in this manuscript are built from cached processed data and public acquisition metadata. Table 1 is generated from the synthetic paper-asset cache and supports CLM-001 as a descriptive inventory. Table 2 is generated by the real-data matrix and records the public real-data registry, independence unit, event or target source, and verified processed row counts. Neither table converts row counts into independent failure counts.

Industrial limitations are material to interpretation. MetroPT-style telemetry can have many samples but few independent failure intervals. MetroPT2 is valuable as a temporal replication resource, but replication still depends on comparable labels and preprocessing. SCANIA changes the independence unit to vehicles and repairs, Hydraulic Systems changes it to load cy-

Table 1: Dataset and split summary.

dataset	split	rows	failures	regimes
synthetic_cyclic	test	4000	401	3
synthetic_cyclic	train	12000	0	3
synthetic_cyclic	validation	4000	2000	3

Table 2: Registered real-data characteristics generated by the real-data matrix.

Dataset	Version	Unit	Event source	Rows
hydraulic_systems	UCI dataset 447	load cycle	cycle-wise component condition targets in profile.txt	2205
metropt	Zenodo record 6854240	failure episode / operating day, not timestamp	version-controlled failure metadata YAML	10773588
metropt2	Zenodo record 7766691	failure episode / operating day, not timestamp	dataset-specific timestamped failure metadata in code	7116940
scania_component_x	Researchdata.se dataset 2024-34	vehicle	published labels and time-to-event files	1516819
secom	UCI dataset 179	wafer / production example	pass/fail yield labels in secom_labels.data	1567

cles, and SECOM changes it to production examples. Transfer claims must therefore match the dataset-specific estimand. The current generated claim ledger blocks a positive population-level superiority claim for industrial predictive maintenance; this limitation is recorded as CLM-007.

6 Experimental Protocol

The evaluation protocol is frozen before final experiment execution. It declares the train, validation, calibration, and test partitions; the exclusion and maintenance-boundary fields; warning horizons; alarm merge gap; matching tolerance; matching method; bootstrap unit; primary metrics; secondary metrics; and event utility costs. The protocol hash used by this manuscript is 96461a9ff099d0d7.

Primary event metrics are event precision, event recall, event F_1 , false-alarm events per operating day, duplicate alarms per failure, median warning lead time, and horizon Brier score. Secondary metrics include pointwise ranking metrics, time under warning, and event utility. Baselines are required to use the same causal feature availability and the same frozen event conversion, so a model is not rewarded for using future information or a more favorable matching rule.

Uncertainty is attached to the independent unit declared by the dataset and protocol. For simulation cells, the unit is the Monte Carlo repetition or registered replicate. For MetroPT-style case studies, the unit is the failure episode or operating interval, not the timestamp. For SCANIA, the relevant unit is the vehicle or repair history defined by the dataset release. For Hydraulic Systems, it is the load cycle with component-condition labels. For SECOM, it is the wafer or production example with a pass/fail yield outcome. Multiplicity is handled by keeping primary metrics and hypotheses predeclared, reporting dataset-specific estimands, and routing every interpretive statement through the claim ledger.

The registered real-data matrix has three audit layers. The verification layer checks local raw and processed manifests, file hashes, row counts, schema summaries, and declared independence units. The industrial-result layer writes one conservative diagnostic row per registered real dataset. The evidence-scope layer combines both outputs into a machine-readable summary of datasets, workflows, sampling structures, independent units, diagnostic statuses, and claim boundaries. This protocol is intentionally strict: a broader dataset base strengthens external coverage only when the estimand differences remain visible.

7 Results

The generated results support a diagnostic and reproducibility claim, not a broad industrial performance claim. Synthetic figures and tables are produced by version-controlled code under

Table 3: Evidence-scope summary generated from real-data verification and industrial-result artifacts.

Dataset	Sampling structure	Unit	Verification	Industrial result
hydraulic_systems	hydraulic test-rig cycles with multi-rate sensor traces	load cycle	verified	succeeded
metropt	ordered compressor telemetry	failure episode / operating day, not timestamp	verified	succeeded
metropt2	ordered compressor telemetry	failure episode / operating day, not timestamp	verified	succeeded
scania_component_x	fleet-level irregular vehicle histories	vehicle	verified	succeeded
secom	semiconductor process measurements per production entity	wafer / production example	verified	succeeded

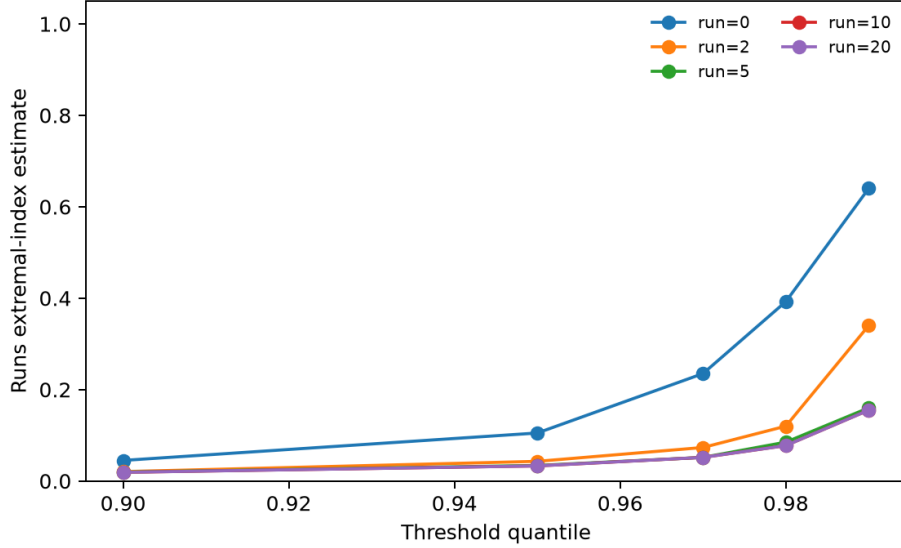


Figure 2: Threshold and run-length stability diagnostic. The figure is generated from the cached processed data and records the empirical runs extremal-index estimate across the registered grid.

experiment identifier 0b597ad010969c41 and dataset checksum 3468c21feca0. Real-data tables are produced by the registered matrix after local public-data acquisition, preparation, manifest verification, and conservative dataset-specific diagnostics.

7.1 Synthetic Diagnostic Assets

Threshold and run-length sensitivity is shown in Figure 2 and Table 4. The grid changes exceedance and cluster behavior, supporting CLM-002. This is expected under dependent telemetry and is the reason the manuscript reports sensitivity rather than a single chosen threshold as if it were model-free.

The alarm frontier in Figure 3 and Table 5 reports the trade-off between event recall, lead time, and false-alarm event rate. It supports CLM-004 only as an exploratory operational trade-off. In particular, a visually acceptable frontier is not evidence of calibrated failure probability, and a reduction in duplicate alarms would be useful only if recall, lead time, and time under warning remain acceptable.

Matching sensitivity is shown in Figure 4 and Table 6. This surface varies both the alarm merge gap and the post-onset matching tolerance, making explicit whether an operational conclusion depends on one favorable episode-conversion rule. It extends CLM-004 but does not change its permitted strength: the evidence is still an exploratory operational trade-off.

Calibration is reported as a reliability diagnostic rather than a probability guarantee. Figure 5 uses ranked horizon-risk surrogates and observed future-event indicators. Figure 6 and Table 7 add split-aware intervals using training-ranked risk scores. This supports CLM-005 as a narrow diagnostic claim: generated intervals are now present, but they are not a substitute for a prospective calibrated probability model.

Table 4: Threshold and run-length stability diagnostics.

threshold_quantile	run_length	n_exceedances	n_clusters
0.9	0	2000	90
0.9	2	2000	41
0.9	5	2000	39
0.9	10	2000	38
0.9	20	2000	38
0.95	0	1000	105
0.95	2	1000	43
0.95	5	1000	34
0.95	10	1000	33
0.95	20	1000	33
0.97	0	600	141
0.97	2	600	44
0.97	5	600	31
0.97	10	600	31
0.97	20	600	31
0.98	0	400	157
0.98	2	400	48
0.98	5	400	34
0.98	10	400	31
0.98	20	400	31
0.99	0	200	128
0.99	2	200	68
0.99	5	200	32
0.99	10	200	31

Table 5: Lead-time versus false-alarm frontier.

threshold_quantile	threshold	event_recall	event_precision	false_alarm_events_per_day	median_warning
0.9	1.119	0	0	168.5	
0.95	1.329	0	0	146.9	
0.97	1.388	0	0	133.9	
0.98	1.411	0	0	146.9	
0.99	1.439	0	0	138.2	

Table 6: Event matching tolerance and alarm merge-gap sensitivity.

merge_gap	matching_tolerance_after	event_recall	event_precision	event_f1	false_alarm_events	duplica
0	0	0	0	0	157	0
0	30	0	0	0	157	0
0	60	0	0	0	157	0
0	120	0	0	0	157	0
0	300	0	0	0	157	0
5	0	0	0	0	34	0
5	30	0	0	0	34	0
5	60	0	0	0	34	0
5	120	0	0	0	34	0
5	300	0	0	0	34	0
20	0	0	0	0	31	0
20	30	0	0	0	31	0
20	60	0	0	0	31	0
20	120	0	0	0	31	0
20	300	0	0	0	31	0
60	0	0	0	0	31	0
60	30	0	0	0	31	0
60	60	0	0	0	31	0
60	120	0	0	0	31	0
60	300	0	0	0	31	0

Table 7: Split-aware calibration intervals using training-ranked risk scores.

split	lower	upper	count	mean_probability	event_rate	event_rate_ci_lower	event_rate_ci_upper
test	0.9	1	4000	1	0.1003	0.09132	0.1099
train	0	0.1	1199	0.05	0	0	0.003194
train	0.1	0.2	1200	0.15	0	0	0.003191
train	0.2	0.3	1201	0.25	0	0	0.003188
train	0.3	0.4	1199	0.35	0	0	0.003194
train	0.4	0.5	1200	0.45	0	0	0.003191
train	0.5	0.6	1201	0.55	0	0	0.003188
train	0.6	0.7	1200	0.65	0	0	0.003191
train	0.7	0.8	1199	0.75	0	0	0.003194
train	0.8	0.9	1200	0.85	0	0	0.003191
train	0.9	1	1201	0.95	0	0	0.003188
validation	0	0.1	215	0.04765	0.2419	0.1895	0.3033
validation	0.1	0.2	234	0.1494	0.2009	0.1545	0.2568
validation	0.2	0.3	219	0.2529	0.2922	0.236	0.3556
validation	0.3	0.4	280	0.352	0.45	0.3928	0.5086
validation	0.4	0.5	292	0.4522	0.2945	0.2452	0.3492
validation	0.5	0.6	283	0.5512	0.4205	0.3644	0.4787
validation	0.6	0.7	366	0.651	0.5383	0.487	0.5887
validation	0.7	0.8	323	0.7488	0.5077	0.4534	0.5618
validation	0.8	0.9	317	0.8504	0.4227	0.3696	0.4777
validation	0.9	1	1471	0.9883	0.8912	0.8743	0.9061

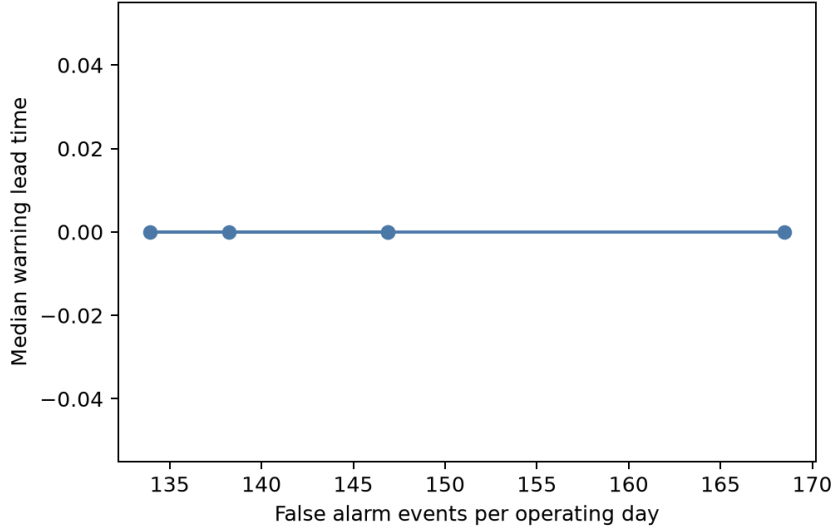


Figure 3: Lead-time versus false-alarm frontier generated under the frozen event-matching protocol.

The ablation registry is summarized in Table 8. It supports CLM-006 as a coverage claim: the artifact records the registered ablation families, but families with missing or non-estimable effect summaries still require direct artifact inspection before a stronger statement is made.

The complete event timeline in Figure 7 is included for auditability. It shows the observable, threshold, labelled failure samples, and alarm points on aligned axes. The figure is useful for detecting visually obvious timing problems, but it is not itself statistical validation.

7.2 Five-Dataset Real-Data Matrix

The real-data matrix verifies five public datasets and records four distinct sampling structures: ordered compressor telemetry, fleet-level irregular vehicle histories, hydraulic test-rig cycles with multi-rate sensor traces, and semiconductor process measurements per production entity. Table 3 shows that the independence units also differ: failure episode or operating day, vehicle, load cycle, and wafer or production example. This is the intended broadening of the study base. It does not make the datasets exchangeable; it makes their non-equivalence auditable.

Table 9 reports the conservative diagnostic row produced for each registered real dataset. MetroPT and MetroPT2 are evaluated as event-level early-warning tasks on temporal test splits. SCANIA is evaluated as vehicle-level repair risk on the published test split. Hydraulic Systems is evaluated as cycle-level hydraulic component degradation, and SECOM as wafer-level semiconductor yield-failure detection. These rows are useful precisely because they preserve weak results. The compressor rows recover the single labelled test event with very low precision, while Hydraulic Systems, SCANIA, and SECOM have $F_1 = 0$ under the registered high-quantile diagnostic. The result is not a performance victory. It is evidence that the pipeline now runs across materially different public industrial-monitoring settings and records when a simple extreme-score diagnostic is insufficient.

The real-data evidence therefore changes the manuscript claim in a specific way. The study is broader than a single synthetic cache or one compressor case study: it now covers five fetched and prepared public datasets with manifest checks, schema summaries, and terminal diagnostic rows. At the same time, CLM-007 remains active because the diagnostic table contains heterogeneous estimands and weak rows, not a controlled demonstration of population-level superiority.

Table 8: Ablation matrix summary.

ablation_family	setting	summary_value
threshold	0.9	0.1
threshold	0.95	0.05
threshold	0.97	0.03
threshold	0.98	0.02
threshold	0.99	0.01
run_length	0	157
run_length	2	48
run_length	5	34
run_length	10	31
run_length	20	31
lag	0	
lag	1	
lag	2	
lag	5	
lag	10	
regime	off	
regime	loaded	
regime	recovery	
dangerous_region	prototype	
dangerous_region	constraint	
dangerous_region	density	
noise_missingness	none	
noise_missingness	noise_0.05	
noise_missingness	missing_0.05	

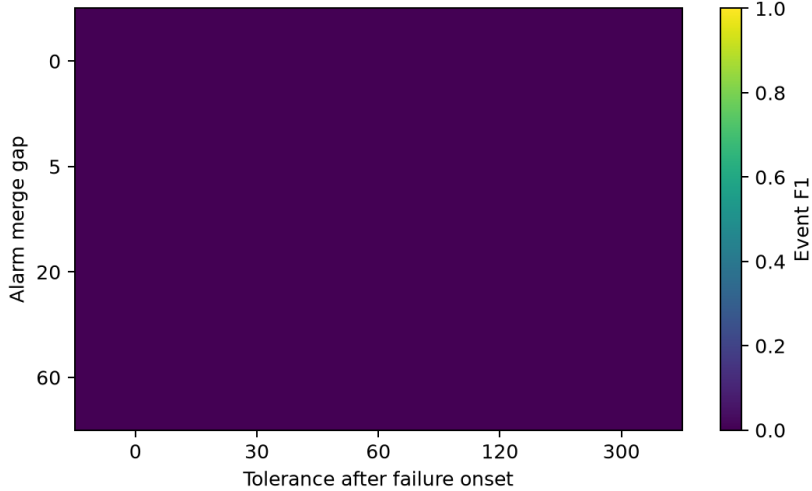


Figure 4: Event F1 sensitivity over alarm merge gap and matching tolerance after failure onset.

Table 9: Conservative real-data diagnostic rows generated by the five-dataset matrix.

Dataset	Status	Estimand	Units	Targets	Predictions	F1
hydraulic_systems	succeeded	cycle-level hydraulic component degradation on chronological test split	441	431	0	0.000
metropt	succeeded	event-level early warning on temporal test split	1	1	2527	0.001
metropt2	succeeded	event-level early warning on temporal test split	1	1	849	0.002
scania_component_x	succeeded	vehicle-level repair risk on published test split	5045	142	132	0.000
secom	succeeded	wafer-level semiconductor yield-failure detection on chronological test split	314	17	10	0.000

8 Discussion

The main value of the workflow is that it makes several failure modes explicit. A low extremal-index estimate may indicate clustered extremes under the chosen threshold and run length, but it can also result from periodicity, persistent operating regimes, preprocessing, missingness, or finite-sample noise. Dynamical EVT supplies the language linking recurrence and extremes, yet the industrial interpretation still requires domain evidence. This paper therefore treats θ as a dependence diagnostic and not as a direct mechanical degradation variable.

Event-level evaluation changes the operational reading of anomaly scores. Pointwise score improvements can be unhelpful if they fragment one failure into many duplicate alarms or if they achieve recall only by keeping a machine under warning for a large fraction of exposure. Conversely, an apparently sparse alarm policy can be unacceptable if it fires too late for maintenance action. The generated frontier and timeline make these trade-offs visible, but they do not remove the need for cost, staffing, and repair-window assumptions.

The five-dataset matrix changes the evidential status of the package. The repository is no longer only a synthetic workflow plus one compressor-family case study. It now verifies public raw and processed artifacts for compressor telemetry, truck repair histories, hydraulic test-rig cycles, and semiconductor process examples, and it writes one terminal diagnostic row for each. This broader base is valuable because it forces the workflow to handle different schemas, labels, missingness patterns, entity structures, and independence units.

The negative result in CLM-007 is still important. The current generated assets do not support broad industrial performance claims beyond the evaluated evidence. That statement is not a weakness to hide; it is a boundary condition for honest use of the artifacts. MetroPT and MetroPT2 have limited independent failure evidence; SCANIA changes the target estimand to vehicle-level risk stratification; Hydraulic Systems uses laboratory cycle-level component states; and SECOM uses wafer-level yield labels with missing high-dimensional process measurements. Future empirical work should therefore add additional public or prospective datasets and pre-

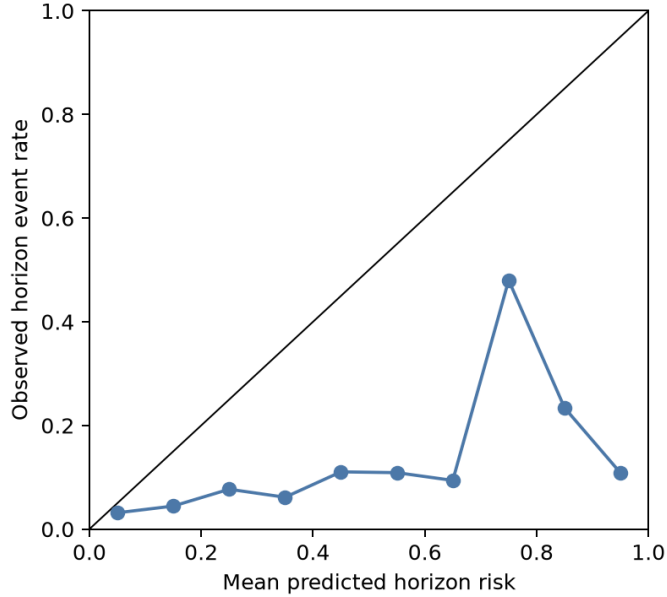


Figure 5: Reliability diagnostic for ranked horizon risk. Bin counts are preserved in the generated table and provenance file.

declared baselines before making a broader predictive-maintenance claim.

The reproducibility layer is also a scientific constraint. Every generated figure and table carries provenance, the claim ledger blocks unsupported manuscript injection, and the paper-check command audits citations, claims, generated assets, and compiled PDFs. This structure does not guarantee that every modelling choice is optimal, but it makes unsupported interpretation harder to introduce silently.

9 Conclusion

This paper presents a reproducible, regime-conditioned dynamical-EVT diagnostic workflow for predictive maintenance and condition monitoring. The supported contribution is broader than the initial cache-only version but still bounded: generated artifacts connect causal machine-state observables, clustered extremes, alarm episodes, sensitivity analyses, calibration diagnostics, claim constraints, and a five-dataset real-data evidence matrix. The expanded matrix covers compressor telemetry, fleet repair histories, hydraulic component-condition cycles, and semiconductor process/yield examples, with terminal verification and conservative diagnostic rows for each. The evidence does not support causal mechanical diagnosis, population-level superiority, or validity beyond the evaluated settings. Those claims require additional datasets, prospective deployment evidence, and stronger predeclared comparative baselines under the same frozen protocol.

References

- [1] Romain Aimino, Ana Cristina Moreira Freitas, Jorge Milhazes Freitas, and Mike Todd. Multivariate extreme values for dynamical systems. *arXiv preprint*, 2024.
- [2] Davide Faranda, Jorge Milhazes Freitas, Valerio Lucarini, Giorgio Turchetti, and Sandro Vaienti. Extreme value statistics for dynamical systems with noise. *Nonlinearity*, 26(9): 2597–2622, 2013. doi: 10.1088/0951-7715/26/9/2597.

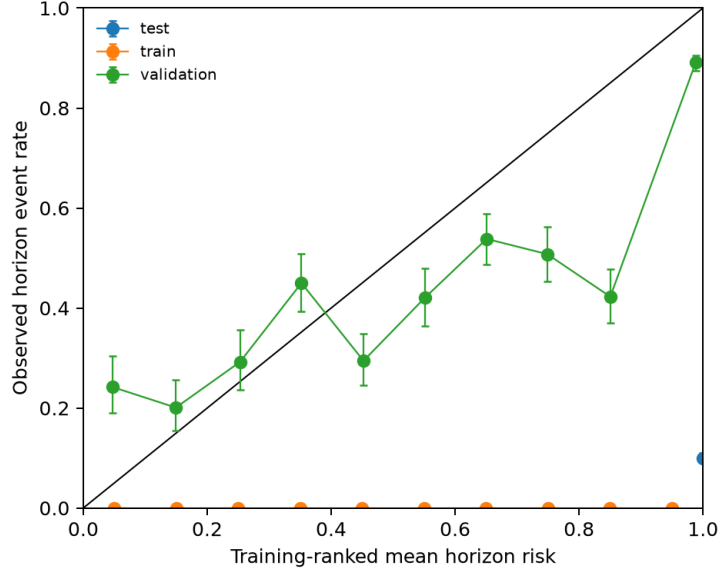


Figure 6: Split-aware calibration intervals for training-ranked horizon-risk scores.

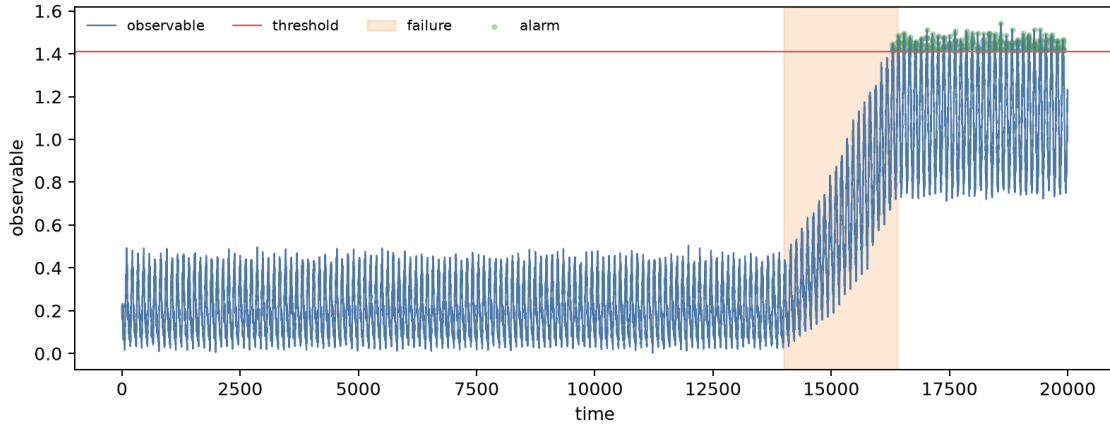


Figure 7: Generated alarm timeline with failures and threshold exceedances on aligned axes.

- [3] Christopher A. T. Ferro and Johan Segers. Inference for clusters of extreme values. *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, 65(2):545–556, 2003. doi: 10.1111/1467-9868.00401.
- [4] Ana Cristina Moreira Freitas, Jorge Milhazes Freitas, and Mike Todd. Hitting time statistics and extreme value theory. *Probability Theory and Related Fields*, 147(3–4):675–710, 2010.
- [5] Ana Cristina Moreira Freitas, Jorge Milhazes Freitas, and Mike Todd. The extremal index, hitting time statistics and periodicity. *Advances in Mathematics*, 231(5):2626–2665, 2012.
- [6] Nikolai Helwig, Eliseo Pignanelli, and Andreas Schütze. Condition monitoring of hydraulic systems, 2018.
- [7] Zahra Kharazian, Tony Lindgren, Sindri Magnússon, Olof Steinert, and Oskar Andersson Reyna. Scania component x dataset: A real-world multivariate time series dataset for predictive maintenance. *Scientific Data*, 12(493), 2025. doi: 10.1038/s41597-025-04802-6.

- [8] Michael McCann and Adrian Johnston. Secom, 2008.
- [9] Alban Siffer, Pierre-Alain Fouque, Alexandre Termier, and Christine Largouët. Anomaly detection in streams with extreme value theory. In *Proceedings of the 23rd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pages 1067–1075. ACM, 2017. doi: 10.1145/3097983.3098144.
- [10] Bruno Veloso, Rita P. Ribeiro, João Gama, and Pedro Mota Pereira. The metropt dataset for predictive maintenance. *Scientific Data*, 9(764), 2022. doi: 10.1038/s41597-022-01877-3.
- [11] Bruno Veloso, Rita P. Ribeiro, Pedro M. Pereira, and João Gama. Metropt2: A benchmark dataset for predictive maintenance, 2023.